Lenses and image formation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use ray directions to explain converging and diverging lens behavior.

SUCCESS CRITERIA - I can...

1. Distinguish converging and diverging lenses.
2. Trace principal rays with a clear optical axis.
3. State whether an image is real or virtual in the given case.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “lens” and “focal point” to explain your model or result. Sentence starters are optional.

lens: a transparent element that changes light-ray directions

focal point: where suitable parallel rays meet or appear to originate

ray: a line showing the modeled direction light travels

01 Bring back an earlier idea

A siren emits 500 Hz steadily while approaching a stationary listener.
Compare observed and emitted frequency and explain.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $1/f = 1/d_o + 1/d_i$.
Thin lens: f is focal length; o = object, i = image. Use the stated sign convention and a ray diagram.

Worked reasoning: A converging lens can form a real image of an object beyond its focal point. A ray parallel to the axis passes through the far focus; a central ray is approximately undeviated in the thin-lens model.

Your turn: Draw these two rays and mark their intersection for an object beyond 2 f.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

Complete ray sketches for parallel rays through a converging lens and a diverging lens.
Label the optical axis.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Left: converging lens. Right: diverging lens. Horizontal lines mark the optical axes.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Build the solution

A supplied converging lens diagram puts an object farther than 2 f away.
Predict the image: upright or inverted; real or virtual.
Use a simple drawing.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

05 Use the idea

A magnifying glass produces an upright virtual image when object is inside f.
Explain why “converging lens always makes a real image”is wrong.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: 1 2 3

06 Choose a method

A wave has period 4 s.
Find the cycles completed in 24 s and its frequency.

Word help: frequency = number of cycles each second

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

Which lens spreads parallel rays: converging or diverging?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Move the object between the converging lens and its focal point.
Predict whether outgoing rays meet physically beyond the lens and describe the image qualitatively.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3